Mahmoud Elbatouty

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EDUCATION

Ph.D. in **Computing, Intelligent Systems (Artificial Intelligence)**, University of North Florida **Anticipated May 2028**
M.S. in **Computing and Information Sciences, Data Science**, University of North Florida — GPA: 4.0/4.0 **May 2025**
B.S. in **Computing and Information Sciences, Data Science**, University of North Florida — GPA: 3.94/4.0 **May 2023**
Minor in Statistics • Summa Cum Laude • Hicks Honors College • Dean's List

PROFESSIONAL EXPERIENCE

Data Scientist / Software Developer — Florida Blue **May 2023 – Present**
Jacksonville, FL

- Engineered an unsupervised clustering pipeline grouping **50+** health-equity dimensions into **3** distinct clusters with **K-Means** after benchmarking against hierarchical and density-based algorithms in **R & Python**; applied **PCA** for dimension reduction, achieving high intra-cluster correlation. Still drives the company's health equity initiatives today.
- Refactored **20+ Python** and **R** codebases into curated internal libraries (visualization templates, **dplyr** wrappers, reusable utilities) adopted across Florida Blue, eliminating repetitive manual work.
- Migrated **120+** processes, jobs, and reports onto a consolidated Microsoft reporting server using **Python**, **Git**, and **Jenkins**, enabling the Apps & Reporting team to decommission legacy servers and significantly reduce operating spend.
- Built an end-to-end automated financial reporting protocol, extracting data from **Microsoft SQL Server** via tailored stored procedures and surfacing per-employee hours and variances by cost-center on a secured **Power BI** dashboard for managers.
- Developed **50+** executive dashboards using **React, JavaScript, Python, SQL, and Power BI** consumed directly by the CEO to monitor financial performance across health-insurance sectors; authored stored procedures computing actuals, budget, and variance.
- Contributed to enterprise cloud initiatives, migrating tables, stored procedures, and security protocols from **SQL Server** to **Microsoft Fabric** and **Snowflake** using **Spark** and on-prem generative AI pipelines.

Modeling and Analytics Engineer — NLP Logix **Sep 2022 – Dec 2022**
Jacksonville, FL

- Automated web-scraping of **Census** data using **Selenium** and **BeautifulSoup**, structuring outputs into relational tables to drive downstream analytical insights.
- Built **14+ machine learning** models in **Python** (**pandas**, **Scikit-Learn**) to forecast child and maternal health outcomes across Florida, surfacing results in **Tableau** dashboards for public-health stakeholders.

Data Scientist — Florida Data Science for Social Good **Jun 2022 – Aug 2022**
Jacksonville, FL

- Performed statistical analysis and data mining on **15 million+** Florida voter records, developing a custom voter-profiling system classifying behavioral patterns for every registered voter in the state from 2014-2020
- Designed and presented **Tableau** dashboards at the Big Reveal to client stakeholders and audience; work contributed to a peer-reviewed publication (below).

PROJECTS

Tax Certificate & Heirs' Property Prediction — LISC AI4Good Hackathon (**3rd Place**) **Sep 2024**

- Predicted whether tax certificates would be sold or redeemed using **Random Forest**, **Logistic Regression**, and **neural networks**, achieving an **AUC-ROC of 0.80** and emitting calibrated risk probabilities per parcel.
- Model adopted by **LISC Jacksonville** to direct intervention efforts, helping allocate **$70K** and assist **75+ families** in retaining their homes; recognized at the UNF Board of Trustees quarterly meeting.

PUBLICATIONS

Johnson, J., Rangarajan, A., Hunt, T., Agarwal, P., **Elbatouty, M.**, Umapathy, K., Richard, D., & Nash, L. (2023). *Finding Insights in Florida Voter Participation.* In *Proceedings of the 2023 International Conference on Advances in Computing Research (ACR'23)*, Lecture Notes in Networks and Systems, vol. 700, pp. 44–55. Springer, Cham. https://doi.org/10.1007/978-3-031-33743-7_4

TECHNICAL SKILLS

Languages & ML/AI: Python, SQL, R, LLMs & Generative AI, Agentic AI, Java, SAS 9.4, JavaScript, TensorFlow, PyTorch, Scikit-Learn, Keras, RAG, Prompt Engineering, Deep Learning, NLP, Time-Series Forecasting, MLOps
Data & Cloud: Microsoft SQL Server, SSIS, Snowflake, Microsoft Fabric, Spark, AWS, Azure, ETL/ELT, Data Modeling, Stored Procedures. Hive, Hadoop
BI & Tools: Power BI, Tableau, SSRS, React, Plotly, ggplot2, Git, Jenkins, CI/CD, Docker, Jupyter, A/B Testing, Statistical Modeling, Agile/Scrum